Homework 5

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Contents

Problem 1 **2**

Problem 2 **3**

Problem 3 **5**

Problem 4 **7**

Appendix: Toolbox **8**

Problem 1

Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n\to\infty} a_n = +\infty$ if and only if $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

Discussion & Scratch Work

This is an "if and only if" statement, so we prove both directions using the definitions.

Because $a_n > 0$ for all n, the reciprocal $\frac{1}{a_n}$ is well-defined and positive. The key observation is that for positive numbers, taking reciprocals reverses inequalities: for $x > 0$ and $M > 0$,

$$x > M \quad \Longleftrightarrow \quad \frac{1}{x} < \frac{1}{M}.$$

So "a_n eventually exceeds every M" is the same as "$\frac{1}{a_n}$ eventually falls below every ε" (after choosing $\varepsilon = 1/M$ or $M = 1/\varepsilon$).

Proof.
($\Rightarrow$) Assume $\lim_{n\to\infty} a_n = +\infty$. By definition, for every $M > 0$ there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n \geq N$. Let $\varepsilon > 0$ be given and choose $M = 1/\varepsilon$ (note $M > 0$). Then there exists N such that for all $n \geq N$,

$$a_n > \frac{1}{\varepsilon}.$$

Since $a_n > 0$, taking reciprocals yields

$$\frac{1}{a_n} < \varepsilon \qquad \text{for all } n \geq N.$$

Thus for all $n \geq N$ we have $\left|\frac{1}{a_n} - 0\right| = \frac{1}{a_n} < \varepsilon$, which shows $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

($\Leftarrow$) Assume $\lim_{n\to\infty} \frac{1}{a_n} = 0$. By definition, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left|\frac{1}{a_n} - 0\right| < \varepsilon.$$

Because $\frac{1}{a_n} > 0$, this is the same as $\frac{1}{a_n} < \varepsilon$. Now let $M > 0$ be given and choose $\varepsilon = 1/M$ (note $\varepsilon > 0$). Then there exists N such that for all $n \geq N$,

$$\frac{1}{a_n} < \frac{1}{M}.$$

Since $a_n > 0$, taking reciprocals reverses the inequality and gives

$$a_n > M \qquad \text{for all } n \geq N.$$

This is exactly the definition of $\lim_{n\to\infty} a_n = +\infty$. $\square$

Problem 2

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n\to\infty} a_n \leq L \leq \limsup_{n\to\infty} a_n.$$

(Hint: As $\{a_n\}_{n\geq 1}$ is not necessarily bounded, we have to consider the cases $\liminf_{n\to\infty} a_n = -\infty$ and $\limsup_{n\to\infty} a_n = +\infty$. However, the proof is much easier in these cases.)

Discussion & Scratch Work

The core idea is: compare each subsequence term a_{k_n} to the *tail* bounds of the original sequence. Fix a tail starting at N. Then

$$u_N := \inf\{a_m : m \geq N\} \quad \text{and} \quad v_N := \sup\{a_m : m \geq N\}$$

satisfy $u_N \leq a_m \leq v_N$ for every $m \geq N$.

Now use the fact that $k_n \to \infty$ (because $\{k_n\}$ is strictly increasing), so for any fixed N, eventually $k_n \geq N$. Hence for all large n,

$$u_N \leq a_{k_n} \leq v_N.$$

Taking $n \to \infty$ gives $u_N \leq L \leq v_N$ for each N. Finally, let $N \to \infty$: by definition,

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N, \qquad \limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N,$$

and passing to the limit preserves the inequalities. (If $\liminf = -\infty$ or $\limsup = +\infty$, the corresponding inequality is automatic.)

Proof.
Let $\{a_{k_n}\}_{n\geq 1}$ be a subsequence of $\{a_n\}_{n\geq 1}$ and suppose

$$\lim_{n\to\infty} a_{k_n} = L \in \mathbb{R}.$$

Case 1: $\liminf_{n\to\infty} a_n = -\infty$. Then $\liminf_{n\to\infty} a_n \leq L$ holds automatically, since $-\infty \leq L$.
Case 2: $\limsup_{n\to\infty} a_n = +\infty$. Then $L \leq \limsup_{n\to\infty} a_n$ holds automatically, since $L \leq +\infty$.
So assume from now on that $\{a_n\}$ is bounded (equivalently, both $\liminf a_n$ and $\limsup a_n$ are finite real numbers). For each $N \in \mathbb{N}$ define the tail infimum and tail supremum

$$u_N = \inf\{a_m : m \geq N\}, \qquad v_N = \sup\{a_m : m \geq N\}.$$

By definition of infimum and supremum, for every $m \geq N$ we have

$$u_N \leq a_m \leq v_N.$$

Fix $N \in \mathbb{N}$. Because $\{k_n\}$ is strictly increasing, we have $k_n \geq n$ for all n, and in particular there exists n_0 such that $k_n \geq N$ for all $n \geq n_0$. Therefore, for all $n \geq n_0$,

$$u_N \leq a_{k_n} \leq v_N.$$

Now take the limit as $n \to \infty$. Since u_N and v_N are constants (in n) and $a_{k_n} \to L$, we obtain

$$u_N \leq L \leq v_N \qquad \text{for every } N \in \mathbb{N}.$$

Finally, let $N \to \infty$. By definition,

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N, \qquad \limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N.$$

Taking limits in the inequality $u_N \leq L \leq v_N$ gives

$$\liminf_{n\to\infty} a_n \leq L \leq \limsup_{n\to\infty} a_n,$$

as desired. $\square$

Problem 3

Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Discussion & Scratch Work

We work directly from the definition of lim sup via tail suprema. We know that they exist for a_n and b_n because the sequences are bounded. We also know they exist for $a_n + b_n$ because the sum of two bounded sequences is bounded (which I think we prove by considering the definition of bounded; if a_n is bounded, then there exists $M_1 > 0$ such that $|a_n| \leq M$ for all n, and similarly for b_n which is bounded by M_2, so $|a_n + b_n| \leq |a_n| + |b_n| \leq$ by the triangle inequality so so $a_n + b_n$ is bounded by $M_1 + M_2$ which we could call $K \in \mathbb{R}$ s.t. $\forall n \in \mathbb{N}, |a_n + b_n| \leq K$ and thus $a_n + b_n$ is bounded). Anyway, check out these definitions:

Since $\{a_n\}$ and $\{b_n\}$ are bounded, for each $N \in \mathbb{N}$ we define

$$v_N^{(a)} = \sup\{a_n : n \geq N\}, \qquad v_N^{(b)} = \sup\{b_n : n \geq N\},$$

and

$$v_N^{(a+b)} = \sup\{a_n + b_n : n \geq N\}.$$

For every $n \geq N$, we have

$$a_n \leq v_N^{(a)} \quad \text{and} \quad b_n \leq v_N^{(b)}.$$

Adding these inequalities gives

$$a_n + b_n \leq v_N^{(a)} + v_N^{(b)}.$$

Since this holds for all $n \geq N$, the supremum satisfies

$$v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}.$$

Now take limits as $N \to \infty$ and use the definition of lim sup.

(Please don't make us prove this again this was so saucey)

Proof.
Since $\{a_n\}$ and $\{b_n\}$ are bounded sequences, for each $N \in \mathbb{N}$ the sets $\{a_n : n \geq N\}$ and $\{b_n : n \geq N\}$ are nonempty and bounded. Define

$$v_N^{(a)} = \sup\{a_n : n \geq N\}, \qquad v_N^{(b)} = \sup\{b_n : n \geq N\},$$

and

$$v_N^{(a+b)} = \sup\{a_n + b_n : n \geq N\}.$$

For every $n \geq N$, we have

$$a_n \leq v_N^{(a)} \quad \text{and} \quad b_n \leq v_N^{(b)}.$$

Therefore

$$a_n + b_n \leq v_N^{(a)} + v_N^{(b)}.$$

Since this inequality holds for all $n \geq N$, taking the supremum over $n \geq N$ gives

$$v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}.$$

Now take limits as $N \to \infty$. By definition,

$$\limsup_{n\to\infty}(a_n + b_n) = \lim_{N\to\infty} v_N^{(a+b)},$$

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N^{(a)}, \qquad \limsup_{n\to\infty} b_n = \lim_{N\to\infty} v_N^{(b)}.$$

Passing to the limit in the inequality $v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}$ yields

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

$\square$

Problem 4

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that $\limsup_{n\to\infty} |a_n| = 0$ if and only if $\lim_{n\to\infty} a_n = 0$.

Discussion & Scratch Work

This is another "if and only if" statement, so we prove both directions.

A helpful way to think about lim sup is via

$$\limsup_{n\to\infty} x_n = \lim_{N\to\infty} \Big(\sup\{x_n : n \geq N\} \Big)$$

whenever the sequence $\{x_n\}$ is bounded above (and if it is not bounded above, the lim sup is $+\infty$).

Here $x_n = |a_n| \geq 0$ always. So:

- If $a_n \to 0$, then eventually $|a_n|$ is small; consequently every tail supremum $\sup\{|a_n| : n \geq N\}$ must also be small, forcing $\limsup |a_n| = 0$.
- Conversely, if the tail suprema of $|a_n|$ tend to 0, then in particular each term $|a_n|$ in the tail is bounded by something $< \varepsilon$, which is exactly the definition of $a_n \to 0$.

Proof.
($\Rightarrow$) Assume $\lim_{n\to\infty} a_n = 0$. Let $\varepsilon > 0$ be given. By definition of convergence, there exists $N_0 \in \mathbb{N}$ such that

$$|a_n - 0| = |a_n| < \varepsilon \qquad \text{for all } n \geq N_0.$$

Fix any $N \geq N_0$. Then for every $n \geq N$ we have $|a_n| < \varepsilon$, and hence

$$\sup\{|a_n| : n \geq N\} \leq \varepsilon.$$

Define $v_N := \sup\{|a_n| : n \geq N\}$. The inequality above shows that for all $N \geq N_0$, $v_N \leq \varepsilon$. Taking $N \to \infty$ yields

$$\limsup_{n\to\infty} |a_n| = \lim_{N\to\infty} v_N \leq \varepsilon.$$

Because $\varepsilon > 0$ was arbitrary and $\limsup_{n\to\infty} |a_n| \geq 0$ (since $|a_n| \geq 0$), we conclude $\limsup_{n\to\infty} |a_n| = 0$.

($\Leftarrow$) Assume $\limsup_{n\to\infty} |a_n| = 0$. Set again

$$v_N := \sup\{|a_n| : n \geq N\}.$$

By definition of lim sup (and since $|a_n| \geq 0$ is bounded above whenever $\limsup |a_n| = 0$ is finite), we have $\lim_{N\to\infty} v_N = 0$. Let $\varepsilon > 0$ be given. Then there exists $N_0 \in \mathbb{N}$ such that

$$v_{N_0} < \varepsilon.$$

But for every $n \geq N_0$, we have $|a_n| \leq v_{N_0}$ (because v_{N_0} is the supremum of the tail $\{|a_n| : n \geq N_0\}$). Therefore, for all $n \geq N_0$,

$$|a_n| \leq v_{N_0} < \varepsilon.$$

This is exactly the definition of $a_n \to 0$. Hence $\lim_{n\to\infty} a_n = 0$. $\square$

Appendix: Toolbox

Sequences and Cauchy

Definition 1. *We say that $\{a_n\}_{n\geq 1}$ is a Cauchy sequence if: for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \varepsilon$ for all $m, n \geq N$.*

Theorem 1 (Cauchy criterion)**.** *Let $\{a_n\}_{n\geq 1}$ be a sequence. The sequence $\{a_n\}_{n\geq 1}$ is Cauchy if and only if it converges.*

Proposition 1. *If $\{a_n\}_{n\geq 1}$ converges, then it is Cauchy.*

Lemma. *If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it is bounded.*

Proposition 2. *If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it converges.*

Subsequences

Definition 2. *Let $\{a_n\}_{n\geq 1}$ be a sequence. We say $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ if $\{k_n\}_{n\geq 1}$ is a strictly increasing sequence of natural numbers: $k_n \in \mathbb{N}$ and $k_n < k_{n+1}$ for all $n \in \mathbb{N}$.*

Remark. *Any such $\{k_n\}_{n\geq 1}$ must satisfy $k_n \geq n$ for all $n \geq 1$. This is easy to prove by induction. (Exercise.)*

Theorem 2. *Let $\{a_n\}_{n\geq 1}$ be a sequence and $a \in \mathbb{R}$. The sequence $\{a_n\}_{n\geq 1}$ converges to a if and only if every subsequence $\{a_{k_n}\}_{n\geq 1}$ converges to a.*

Monotone Subsequences

Proposition 3. *For any sequence, there is a subsequence that is monotone.*

Theorem 3 (Bolzano–Weierstrass)**.** *For any bounded sequence, there exists a subsequence that converges.*

Divergence to $\pm\infty$

Definition 3. *Let $\{a_n\}_{n\geq 1}$ be a sequence. We say:*

(1) *$\{a_n\}_{n\geq 1}$ diverges to $+\infty$ if for all $M > 0$ there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n \geq N$. When this is true, we write $\lim_{n\to\infty} a_n = +\infty$.*

(2) *$\{a_n\}_{n\geq 1}$ diverges to $-\infty$ if for all $m < 0$ there exists $N \in \mathbb{N}$ such that $a_n < m$ for all $n \geq N$. When this is true, we write $\lim_{n\to\infty} a_n = -\infty$.*

Proposition 4. *If $\{a_n\}_{n\geq 1}$ is unbounded and increasing (decreasing), then $\{a_n\}_{n\geq 1}$ diverges to $+\infty$ ($-\infty$, respectively).*

Remark. *If a sequence is increasing, there are only two possibilities:*

(1) *bounded* $\Rightarrow$ *converges;*

(2) *unbounded* $\Rightarrow$ *diverges to* $+\infty$.

Remark. *"*$\lim a_n$*" only sometimes makes sense:*

(1) *converges* $\Rightarrow \lim_{n\to\infty} a_n = a \in \mathbb{R}$;

(2) *diverges: diverges to* $\pm\infty \Rightarrow \lim_{n\to\infty} a_n = \pm\infty$, *else* $\lim_{n\to\infty} a_n$ *does not exist.*

Limit Superior and Limit Inferior

Definition 4. *Let* $\{a_n\}_{n\geq 1}$ *be a sequence (convergent or not). Case 1:* $\{a_n\}_{n\geq 1}$ *is bounded. Fix any* $N \in \mathbb{N}$. *Then the set* $\{a_n : n \geq N\}$ *is nonempty and bounded, so*

$$u_N = \inf\{a_n : n \geq N\}, \qquad v_N = \sup\{a_n : n \geq N\}$$

both exist. We automatically have $u_N \leq v_N$ *for all* N.

Proposition 5. *With* u_N *and* v_N *as above,* $\{u_N\}_{N\geq 1}$ *is increasing and* $\{v_N\}_{N\geq 1}$ *is decreasing. In particular,*

$$u_1 \leq u_2 \leq \cdots \leq u_N \leq v_N \leq \cdots \leq v_2 \leq v_1 \quad \textit{for all } N \geq 1.$$

Proposition 6. *The sequences* $\{u_N\}_{N\geq 1}$ *and* $\{v_N\}_{N\geq 1}$ *are both monotone and bounded, and therefore converge. Set*

$$u = \lim_{N\to\infty} u_N \qquad \textit{and} \qquad v = \lim_{N\to\infty} v_N.$$

Then $u \leq v$. *Moreover, if* $\lim_{n\to\infty} a_n = a$ *exists, then* $u \leq a \leq v$.

Definition 5. *Let* $\{a_n\}_{n\geq 1}$ *be a bounded sequence. We define*

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N = \lim_{N\to\infty} \Big(\sup\{a_n : n \geq N\}\Big)$$

and

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N = \lim_{N\to\infty} \Big(\inf\{a_n : n \geq N\}\Big).$$

Corollary 1. *If* $\{a_n\}_{n\geq 1}$ *is bounded, then*

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

Definition 6. *If* $\{a_n\}_{n\geq 1}$ *is not bounded above, then we define*

$$\limsup_{n\to\infty} a_n = +\infty.$$

If $\{a_n\}_{n\geq 1}$ *is not bounded below, then we define*

$$\liminf_{n\to\infty} a_n = -\infty.$$

Remark. *So* $\limsup$ *and* $\liminf$ *make sense for any sequence, but their values are in* $\mathbb{R} \cup \{\pm\infty\}$.

Corollary 2. *For any sequence* $\{a_n\}_{n\geq 1}$, *we have*

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

Theorem 4. *Let* $\{a_n\}_{n\geq 1}$ *be a sequence.*

(1) *If* $\lim_{n\to\infty} a_n$ *exists in* $\mathbb{R} \cup \{\pm\infty\}$, *then*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n.$$

(2) *If* $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$ *in* $\mathbb{R} \cup \{\pm\infty\}$, *then* $\lim_{n\to\infty} a_n$ *exists and equals this value.*

Remark. *Here, "*$\lim_{n\to\infty} a_n$ *exists in* $\mathbb{R}\cup\{\pm\infty\}$*" just means "*$\{a_n\}_{n\geq 1}$ *converges, or diverges to* $+\infty$, *or diverges to* $-\infty$*."*